

**REAL ESTATE
MATH
CHEAT SHEET**

Percentages

Converting Percentages to Decimals

$$\text{Six percent} = 6.0\% = 0.\overset{\text{.00.\%}}{\underset{\curvearrowright}{06}}.0\% = 0.06$$

$$40.0\% = 0.\underset{\curvearrowright}{40}.0\% = 0.40$$

$$3.5\% = 0.\underset{\curvearrowright}{03}5\% = 0.035$$

Converting Decimals to Percentages

$$0.045 = 0.\underset{\curvearrowright}{04}.5\% = 4.5\%$$

$$1.00 = 1.\underset{\curvearrowright}{00}\% = 100\%$$

Converting Fractions to Decimals

$$2/3 = 2 \div 3 = 0.667 = 0.\underset{\curvearrowright}{66}.7\% = 66.7\%$$

$$1/10 = 1 \div 10 = 0.10 = 0.\underset{\curvearrowright}{10}\% = 10\%$$

Commissions

Sales price × commission rate = total commission

You just sold a home for \$300,000. The commission

rate is 5%. What is the total commission paid?

Sales price = \$300,000

Commission rate = 5% (or 0.05)

Total commission = ?

Sales price × commission rate = **total commission**

\$300,000 × 0.05 (or 5%) = **\$15,000**

Assume you split the commission with the real estate firm. You keep 60% and the firm gets 40%.

How much do you keep?

\$15,000 × 0.6 = **\$9,000** commission you keep

Total commission ÷ sales price = commission rate

A commission of \$6,000 was paid on a \$120,000 home. What is the commission rate?

Sales price = \$120,000

Commission rate = ?

Total commission = \$6,000

Total commission ÷ sales price = **commission rate**

\$6,000 ÷ \$120,000 = **0.05 (or 5%)**

Total commission ÷ commission rate = sales price

What was the sale price on a property if a 4.5% commission rate paid a total commission of \$18,000?

Sales price = ?

Commission rate = 4.5% (or 0.045)

Total commission = \$18,000

$$\begin{array}{rclclcl} \text{Total commission} & \div & \text{commission rate} & = & \text{sales price} \\ \$18,000 & \div & 0.045 & = & \mathbf{\$400,000} \end{array}$$

Solving for Variables

With any two numbers in a formula, you can solve for the third.

Sales price × commission rate = total commission

$$? \times 5.5\% = \$11,000$$

$$? \times 5.5\% = \$11,000 \div 5.5\%$$

$$? = \$11,000 \div .055$$

$$? = \mathbf{\$200,000}$$

You can figure out the sales price of a home with just the commission rate and the commission paid.

Likewise, you could figure out the commission rate with the sales price and commission paid.

Net to Seller

$$\text{Sales price} \times (100\% - \% \text{ commission}) = \text{net to seller}$$

A home just sold for \$400,000. What will the seller net after the 7% commission is deducted?

$$\text{Sales price} \times (100\% - \% \text{ commission}) = \text{net to seller}$$

$$\$400,000 \times (100 - 7\%) = \text{net to seller}$$

$$400,000 \times 93\% = \text{net to seller}$$

$$400,000 \times 0.93 = \text{\$372,000}$$

Loan Interest

$$\text{Loan balance} \times \text{interest rate} = \text{annual interest (\$)}$$

How much will a homeowner pay annually in interest on a \$275,000 loan with a 4.75% interest rate?

$$\text{Loan balance} = \$275,000$$

$$\text{Interest rate} = 4.75\% \text{ (or } 0.0475\text{)}$$

$$\text{Annual interest payment} = ?$$

$$\text{Loan balance} \times \text{interest rate} = \text{annual interest (\$)}$$

$$\$275,000 \times 0.0475 = \text{\$13,062.50}$$

$$\text{Annual interest} \div \text{loan balance} = \text{interest rate}$$

What's the interest rate if \$12,000 is paid in annual interest on a loan balance of \$455,000?

$$\text{Loan balance} = \$455,000$$

Interest rate = ?

Annual interest payment = \$12,000

Annual interest $\div$ loan balance = **interest rate**

$$\$12,000 \div \$455,000 = \mathbf{0.0264 \text{ (or } 2.64\%)}$$

Annual interest $\div$ interest rate = loan balance

What is the current loan balance if \$15,000 is paid toward interest on a loan with a 7.5% rate?

Loan balance = ?

Interest rate = 7.5% (or 0.075)

Annual interest payment = \$15,000

Annual interest $\div$ interest rate = **loan balance**

$$\$15,000 \div 0.075 = \mathbf{\$200,000}$$

Points

1 point = 1% loan amount

Loan amount $\times$ points (%) = points in dollars

A buyer must pay four points at closing on a \$100,000 home loan. How much is this?

4 points = 4% = 0.04

$$\$100,000 \times 0.04 = \mathbf{\$4,000}$$

Appreciation

**Original value × appreciation rate =
amount of appreciation (\$)**

A property valued at \$225,000 today is expected to appreciate 2% each year. By how much would the property's value increase after one year?

$$\$225,000 \times 0.02 = \$4,500$$

What is the new property value?

$$\$225,000 + \$4,500 = \$229,500$$

*What will the property's value be after **two** years?*

$$\$229,500 \times 0.02 = \$4,590$$

$$\$229,500 + \$4,590 = \$234,090$$

Depreciation

**Original value × value not lost (%) =
depreciated value**

Market factors indicate that a new parcel of land valued at \$40,000 is going to decrease in value by 3.5% over the next year. What will its new value be after a year?

$$100\% - 3.5\% = 96.5\%$$

$$\$40,000 \times 0.965 = \$38,600$$

Basic Mortgage Qualifying

Loan amount × origination % = origination fee

A 1% loan origination fee will be due at closing on a \$125,000 loan. What amount is this?

$$\$125,000 \times 0.01 = \text{\textbf{\$1,250 loan origination fee}}$$

Loan amount ÷ home value = loan-to-value ratio

An institution will lend up to \$112,500 for a property appraised at \$125,000. What is the loan-to-value ratio?

$$\$112,500 \div \$125,000 = .90 \text{ or } \text{\textbf{90\% LTV}}$$

Monthly debt obligations ÷ monthly gross income = total debt-to-income ratio

A consumer has total monthly debt obligations of \$1,000 and a gross income of \$3,500. What is the debt-to-income ratio?

$$\$1,000 \div \$3,500 = 0.286 \text{ or } \text{\textbf{28.6\% total DTI}}$$

Monthly housing expenses ÷ monthly gross income = housing ratio

A consumer anticipates housing-related expenses of \$1,500 and has a gross income of \$4,000. What is the

housing ratio?

$$\$1,500 \div \$4,000 = 0.375 \text{ or } \mathbf{37.5\% \text{ housing ratio}}$$

Rates

**Mortgage amount $\times$ tax rate =
mortgage recording tax**

- ! This tax may occur at the state, county, and/or local level and can be made up of separate taxes added together.

The mortgage recording tax rate for a \$325,000 home is \$2.75 per \$100 mortgaged. If the principal amount of the mortgage is \$200,000, what is the mortgage recording tax amount?

$$\text{Rate} = \$2.75 \div \$100 = 0.0275$$

$$\$200,000 \times 0.0275 = \mathbf{\$5,500}$$

Assessed property value $\times$ tax rate = property tax (annual)

If a home's assessed value is \$325,000 and the tax rate is 1.25%, what is the annual property tax amount?

$$\$325,000 \times 0.0125 = \mathbf{\$4,062.50}$$

Assessed property value $\div$ 1,000 = per mill (\$)

A property assessed at \$80,000 pays a tax rate of 100 mills. What does the owner pay in taxes?

$$\$80,000 \div 1,000 = \$80 \text{ per mill}$$

$$\$80 \times 100 \text{ mills} = \text{\textbf{\$8,000 in annual taxes}}$$

$$\text{\textbf{Assessed property value}} \times \text{\textbf{mill rate}} \div 1,000 = \text{\textbf{annual taxes (\$)}}$$